

Table 6: Direct answer and CoT accuracies for each reasoning category across models.

Model	Commonsense		Knowledge		Mathematical		Symbolic		Soft	
	DA %	CoT %	DA %	CoT %	DA %	CoT %	DA %	CoT %	DA %	CoT %
Claude-3 Haiku	74.3	77.2	73.0	76.1	18.1	48.2	38.6	48.7	55.9	56.6
Claude-3.5 Sonnet	84.3	85.8	83.8	88.8	38.7	59.0	53.2	67.1	67.6	75.7
GPT-4o Mini	81.8	83.2	73.6	83.1	22.9	59.7	48.1	60.9	61.1	63.5
Gemini 1.5 Flash	80.3	76.8	78.2	81.0	27.2	55.7	47.0	59.7	60.6	62.6
Gemini 1.5 Pro	80.4	78.3	80.9	83.8	35.4	58.5	52.9	62.6	64.1	67.8
Gemma 2 9b	75.0	76.1	74.9	76.9	18.5	50.5	46.7	55.8	58.2	60.5
Gpt-4o	87.3	87.7	82.9	88.6	36.5	63.3	55.7	68.3	65.9	74.0
Meta-Llama 2 7b	51.4	50.9	44.1	46.6	9.3	17.2	22.4	35.4	37.2	37.6
Meta-Llama 3.1 70b	84.2	84.7	82.4	85.6	24.9	54.9	49.0	60.0	65.7	69.5
Meta-Llama 3.1 8b	72.9	73.4	70.1	74.1	16.0	47.8	34.8	51.6	55.0	56.2
Mistral 7b	58.3	61.8	62.0	64.5	10.9	28.9	41.8	45.0	48.6	49.7
Phi-3 Small 8k	70.8	72.5	76.1	79.7	17.8	47.1	51.2	58.7	57.9	56.4
Qwen 2 72b	82.9	84.9	78.6	84.6	23.9	58.5	48.2	58.7	64.2	65.1
Qwen 2 7b	64.0	66.1	65.2	71.3	15.9	53.5	43.8	52.3	54.4	49.4
Average	74.8	75.7	73.3	77.5	22.6	50.2	45.2	56.1	58.3	60.3

Table 7: Zero-shot accuracy for direct answering and CoT prompts on all datasets

Dataset	Type	Model	zero-shot CoT accuracy	zero-shot DA accuracy
MuSR Team Allocations	Soft Reasoning	Llama 2 7b	34.8	37.2
MuSR Team Allocations	Soft Reasoning	Mistral 7b	38.8	46.8
MuSR Team Allocations	Soft Reasoning	Llama 3.1 8b	44.0	48.0
MuSR Team Allocations	Soft Reasoning	Llama 3.1 70b	65.2	66.8
MuSR Team Allocations	Soft Reasoning	Gemma 2 9b	47.2	44.8
MuSR Team Allocations	Soft Reasoning	Phi-3 Small 8k	47.2	61.6
MuSR Team Allocations	Soft Reasoning	Qwen 2 7b	42.0	49.6
MuSR Team Allocations	Soft Reasoning	Qwen 2 72b	58.0	66.8
MuSR Team Allocations	Soft Reasoning	GPT-4o Mini	61.2	58.4
MuSR Team Allocations	Soft Reasoning	Gpt-4o	64.0	63.6
MuSR Team Allocations	Soft Reasoning	Claude-3 Haiku	56.8	59.2
MuSR Team Allocations	Soft Reasoning	Claude-3.5 Sonnet	80.4	63.2
MuSR Team Allocations	Soft Reasoning	Gemini 1.5 Flash	48.8	55.2
MuSR Team Allocations	Soft Reasoning	Gemini 1.5 Pro	58.4	62.4
SiQA	Commonsense	Llama 2 7b	53.4	55.9
SiQA	Commonsense	Mistral 7b	35.9	33.5
SiQA	Commonsense	Llama 3.1 8b	73.5	73.5
SiQA	Commonsense	Llama 3.1 70b	78.7	80.9
SiQA	Commonsense	Gemma 2 9b	74.9	76.3
SiQA	Commonsense	Phi-3 Small 8k	38.0	40.4
SiQA	Commonsense	Qwen 2 7b	37.3	39.3
SiQA	Commonsense	Qwen 2 72b	80.5	80.4
SiQA	Commonsense	GPT-4o Mini	79.0	80.0
SiQA	Commonsense	Gpt-4o	81.9	81.5
SiQA	Commonsense	Claude-3 Haiku	75.4	74.8
SiQA	Commonsense	Claude-3.5 Sonnet	79.7	81.0
SiQA	Commonsense	Gemini 1.5 Flash	74.5	79.1
SiQA	Commonsense	Gemini 1.5 Pro	73.9	78.2
MuSiQue	Soft Reasoning	Llama 2 7b	40.1	36.1
MuSiQue	Soft Reasoning	Mistral 7b	47.3	47.2
MuSiQue	Soft Reasoning	Llama 3.1 8b	62.6	64.7
MuSiQue	Soft Reasoning	Llama 3.1 70b	74.0	72.2
MuSiQue	Soft Reasoning	Gemma 2 9b	67.7	68.7
MuSiQue	Soft Reasoning	Phi-3 Small 8k	58.3	64.3
MuSiQue	Soft Reasoning	Qwen 2 7b	60.7	65.1
MuSiQue	Soft Reasoning	Qwen 2 72b	56.3	69.0
MuSiQue	Soft Reasoning	GPT-4o Mini	71.3	68.2
MuSiQue	Soft Reasoning	Gpt-4o	73.5	70.1
MuSiQue	Soft Reasoning	Claude-3 Haiku	54.8	56.0
MuSiQue	Soft Reasoning	Claude-3.5 Sonnet	66.9	70.4
MuSiQue	Soft Reasoning	Gemini 1.5 Flash	69.8	66.2
MuSiQue	Soft Reasoning	Gemini 1.5 Pro	69.8	71.3
AGIEval LSAT RC	Soft Reasoning	Llama 2 7b	31.2	36.4
AGIEval LSAT RC	Soft Reasoning	Mistral 7b	61.7	61.0
AGIEval LSAT RC	Soft Reasoning	Llama 3.1 8b	71.0	68.8
AGIEval LSAT RC	Soft Reasoning	Llama 3.1 70b	84.4	87.0
AGIEval LSAT RC	Soft Reasoning	Gemma 2 9b	75.1	78.1
AGIEval LSAT RC	Soft Reasoning	Phi-3 Small 8k	68.8	69.9
AGIEval LSAT RC	Soft Reasoning	Qwen 2 7b	61.0	66.5
AGIEval LSAT RC	Soft Reasoning	Qwen 2 72b	83.6	84.4

Table 7: Zero-shot accuracy for direct answering and CoT prompts on all datasets

Dataset	Type	Model	zero-shot CoT accuracy	zero-shot DA accuracy
AGIEval LSAT RC	Soft Reasoning	GPT-4o Mini	77.3	74.3
AGIEval LSAT RC	Soft Reasoning	Gpt-4o	88.1	81.4
AGIEval LSAT RC	Soft Reasoning	Claude-3 Haiku	71.7	65.1
AGIEval LSAT RC	Soft Reasoning	Claude-3.5 Sonnet	90.0	89.6
AGIEval LSAT RC	Soft Reasoning	Gemini 1.5 Flash	78.1	81.0
AGIEval LSAT RC	Soft Reasoning	Gemini 1.5 Pro	82.2	85.9
CommonsenseQA	Commonsense	Llama 2 7b	49.4	54.6
CommonsenseQA	Commonsense	Mistral 7b	68.0	68.0
CommonsenseQA	Commonsense	Llama 3.1 8b	68.5	74.9
CommonsenseQA	Commonsense	Llama 3.1 70b	83.5	84.4
CommonsenseQA	Commonsense	Gemma 2 9b	79.2	80.1
CommonsenseQA	Commonsense	Phi-3 Small 8k	81.8	80.3
CommonsenseQA	Commonsense	Qwen 2 7b	78.5	79.0
CommonsenseQA	Commonsense	Qwen 2 72b	87.4	87.3
CommonsenseQA	Commonsense	GPT-4o Mini	82.5	83.9
CommonsenseQA	Commonsense	Gpt-4o	86.5	87.3
CommonsenseQA	Commonsense	Claude-3 Haiku	80.6	79.0
CommonsenseQA	Commonsense	Claude-3.5 Sonnet	85.1	84.3
CommonsenseQA	Commonsense	Gemini 1.5 Flash	79.7	82.6
CommonsenseQA	Commonsense	Gemini 1.5 Pro	79.9	82.9
GPQA	Mathematical	Llama 2 7b	28.3	24.3
GPQA	Mathematical	Mistral 7b	23.0	24.3
GPQA	Mathematical	Llama 3.1 8b	24.1	25.9
GPQA	Mathematical	Llama 3.1 70b	23.2	25.9
GPQA	Mathematical	Gemma 2 9b	26.3	21.2
GPQA	Mathematical	Phi-3 Small 8k	22.3	20.8
GPQA	Mathematical	Qwen 2 7b	24.1	24.6
GPQA	Mathematical	Qwen 2 72b	21.0	18.1
GPQA	Mathematical	GPT-4o Mini	21.0	24.0
GPQA	Mathematical	Gpt-4o	23.7	25.9
GPQA	Mathematical	Claude-3 Haiku	25.4	22.3
GPQA	Mathematical	Claude-3.5 Sonnet	25.4	25.9
GPQA	Mathematical	Gemini 1.5 Flash	22.3	22.8
GPQA	Mathematical	Gemini 1.5 Pro	21.0	23.7
AGIEval LSAT LR	Soft Reasoning	Llama 2 7b	29.4	33.5
AGIEval LSAT LR	Soft Reasoning	Mistral 7b	44.1	47.8
AGIEval LSAT LR	Soft Reasoning	Llama 3.1 8b	59.0	53.9
AGIEval LSAT LR	Soft Reasoning	Llama 3.1 70b	81.4	81.0
AGIEval LSAT LR	Soft Reasoning	Gemma 2 9b	64.9	67.6
AGIEval LSAT LR	Soft Reasoning	Phi-3 Small 8k	64.5	64.1
AGIEval LSAT LR	Soft Reasoning	Qwen 2 7b	50.6	58.4
AGIEval LSAT LR	Soft Reasoning	Qwen 2 72b	77.3	75.1
AGIEval LSAT LR	Soft Reasoning	GPT-4o Mini	65.3	68.2
AGIEval LSAT LR	Soft Reasoning	Gpt-4o	87.3	83.9
AGIEval LSAT LR	Soft Reasoning	Claude-3 Haiku	55.7	54.7
AGIEval LSAT LR	Soft Reasoning	Claude-3.5 Sonnet	83.7	82.7
AGIEval LSAT LR	Soft Reasoning	Gemini 1.5 Flash	70.0	71.2
AGIEval LSAT LR	Soft Reasoning	Gemini 1.5 Pro	79.4	80.4
PiQA	Commonsense	Llama 2 7b	62.1	64.7
PiQA	Commonsense	Mistral 7b	78.6	77.7
PiQA	Commonsense	Llama 3.1 8b	85.0	84.2
PiQA	Commonsense	Llama 3.1 70b	91.8	90.6
PiQA	Commonsense	Gemma 2 9b	84.0	84.8
PiQA	Commonsense	Phi-3 Small 8k	89.1	85.5
PiQA	Commonsense	Qwen 2 7b	84.3	86.2
PiQA	Commonsense	Qwen 2 72b	92.9	89.1
PiQA	Commonsense	GPT-4o Mini	93.1	88.6
PiQA	Commonsense	Gpt-4o	95.9	95.5
PiQA	Commonsense	Claude-3 Haiku	85.9	86.6
PiQA	Commonsense	Claude-3.5 Sonnet	94.6	94.5
PiQA	Commonsense	Gemini 1.5 Flash	84.6	89.8
PiQA	Commonsense	Gemini 1.5 Pro	88.1	91.3
Arc Easy	Knowledge	Llama 2 7b	71.1	69.8
Arc Easy	Knowledge	Mistral 7b	87.5	86.7
Arc Easy	Knowledge	Llama 3.1 8b	93.0	92.5
Arc Easy	Knowledge	Llama 3.1 70b	97.5	97.9
Arc Easy	Knowledge	Gemma 2 9b	94.9	95.8
Arc Easy	Knowledge	Phi-3 Small 8k	96.0	96.3
Arc Easy	Knowledge	Qwen 2 7b	89.5	84.7
Arc Easy	Knowledge	Qwen 2 72b	97.9	97.4
Arc Easy	Knowledge	GPT-4o Mini	96.8	94.6
Arc Easy	Knowledge	Gpt-4o	98.9	98.1
Arc Easy	Knowledge	Claude-3 Haiku	95.1	95.4
Arc Easy	Knowledge	Claude-3.5 Sonnet	98.6	98.4
Arc Easy	Knowledge	Gemini 1.5 Flash	96.8	97.2
Arc Easy	Knowledge	Gemini 1.5 Pro	97.2	94.6
Arc Challenge	Knowledge	Llama 2 7b	49.2	45.2

Table 7: Zero-shot accuracy for direct answering and CoT prompts on all datasets

Dataset	Type	Model	zero-shot CoT accuracy	zero-shot DA accuracy
Arc Challenge	Knowledge	Mistral 7b	78.3	76.6
Arc Challenge	Knowledge	Llama 3.1 8b	86.0	82.6
Arc Challenge	Knowledge	Llama 3.1 70b	95.0	93.6
Arc Challenge	Knowledge	Gemma 2 9b	91.0	89.6
Arc Challenge	Knowledge	Phi-3 Small 8k	91.6	91.0
Arc Challenge	Knowledge	Qwen 2 7b	83.9	75.3
Arc Challenge	Knowledge	Qwen 2 72b	96.3	94.6
Arc Challenge	Knowledge	GPT-4o Mini	93.3	82.6
Arc Challenge	Knowledge	Gpt-4o	96.0	95.3
Arc Challenge	Knowledge	Claude-3 Haiku	89.3	89.3
Arc Challenge	Knowledge	Claude-3.5 Sonnet	96.0	95.3
Arc Challenge	Knowledge	Gemini 1.5 Flash	92.3	93.6
Arc Challenge	Knowledge	Gemini 1.5 Pro	91.6	90.6
AGIEval LSAT AR	Soft Reasoning	Llama 2 7b	17.0	17.4
AGIEval LSAT AR	Soft Reasoning	Mistral 7b	21.7	19.1
AGIEval LSAT AR	Soft Reasoning	Llama 3.1 8b	20.4	26.1
AGIEval LSAT AR	Soft Reasoning	Llama 3.1 70b	32.6	28.7
AGIEval LSAT AR	Soft Reasoning	Gemma 2 9b	24.8	23.0
AGIEval LSAT AR	Soft Reasoning	Phi-3 Small 8k	28.3	26.5
AGIEval LSAT AR	Soft Reasoning	Qwen 2 7b	27.0	23.9
AGIEval LSAT AR	Soft Reasoning	Qwen 2 72b	29.1	28.3
AGIEval LSAT AR	Soft Reasoning	GPT-4o Mini	32.2	23.0
AGIEval LSAT AR	Soft Reasoning	Gpt-4o	37.8	30.0
AGIEval LSAT AR	Soft Reasoning	Claude-3 Haiku	24.8	23.5
AGIEval LSAT AR	Soft Reasoning	Claude-3.5 Sonnet	38.3	33.9
AGIEval LSAT AR	Soft Reasoning	Gemini 1.5 Flash	27.8	27.8
AGIEval LSAT AR	Soft Reasoning	Gemini 1.5 Pro	30.0	31.7
BiGGen Bench	Soft Reasoning	Llama 2 7b	61.6	56.8
BiGGen Bench	Soft Reasoning	Mistral 7b	70.1	68.1
BiGGen Bench	Soft Reasoning	Llama 3.1 8b	66.5	67.7
BiGGen Bench	Soft Reasoning	Llama 3.1 70b	78.9	76.9
BiGGen Bench	Soft Reasoning	Gemma 2 9b	64.7	64.5
BiGGen Bench	Soft Reasoning	Phi-3 Small 8k	69.7	63.0
BiGGen Bench	Soft Reasoning	Qwen 2 7b	46.2	69.9
BiGGen Bench	Soft Reasoning	Qwen 2 72b	74.3	79.9
BiGGen Bench	Soft Reasoning	GPT-4o Mini	70.3	77.7
BiGGen Bench	Soft Reasoning	Gpt-4o	86.0	82.0
BiGGen Bench	Soft Reasoning	Claude-3 Haiku	80.0	80.0
BiGGen Bench	Soft Reasoning	Claude-3.5 Sonnet	91.4	79.3
BiGGen Bench	Soft Reasoning	Gemini 1.5 Flash	73.9	68.5
BiGGen Bench	Soft Reasoning	Gemini 1.5 Pro	78.7	67.1
Winogrande	Commonsense	Llama 2 7b	49.9	50.4
Winogrande	Commonsense	Mistral 7b	60.4	56.5
Winogrande	Commonsense	Llama 3.1 8b	66.5	63.3
Winogrande	Commonsense	Llama 3.1 70b	84.2	81.2
Winogrande	Commonsense	Gemma 2 9b	68.7	67.7
Winogrande	Commonsense	Phi-3 Small 8k	81.5	81.6
Winogrande	Commonsense	Qwen 2 7b	67.1	60.7
Winogrande	Commonsense	Qwen 2 72b	81.9	80.7
Winogrande	Commonsense	GPT-4o Mini	79.2	71.9
Winogrande	Commonsense	Gpt-4o	89.7	86.5
Winogrande	Commonsense	Claude-3 Haiku	70.7	66.2
Winogrande	Commonsense	Claude-3.5 Sonnet	89.4	85.7
Winogrande	Commonsense	Gemini 1.5 Flash	72.5	74.8
Winogrande	Commonsense	Gemini 1.5 Pro	75.5	78.3
MMLU	Knowledge	Llama 2 7b	46.3	41.7
MMLU	Knowledge	Mistral 7b	60.5	56.5
MMLU	Knowledge	Llama 3.1 8b	72.6	67.5
MMLU	Knowledge	Llama 3.1 70b	85.0	83.2
MMLU	Knowledge	Gemma 2 9b	73.8	71.4
MMLU	Knowledge	Phi-3 Small 8k	76.3	73.6
MMLU	Knowledge	Qwen 2 7b	67.0	64.5
MMLU	Knowledge	Qwen 2 72b	81.3	77.8
MMLU	Knowledge	GPT-4o Mini	79.9	74.8
MMLU	Knowledge	Gpt-4o	87.5	83.4
MMLU	Knowledge	Claude-3 Haiku	72.2	68.4
MMLU	Knowledge	Claude-3.5 Sonnet	87.2	84.0
MMLU	Knowledge	Gemini 1.5 Flash	76.3	74.7
MMLU	Knowledge	Gemini 1.5 Pro	81.3	81.1
StrategyQA	Commonsense	Llama 2 7b	39.5	31.2
StrategyQA	Commonsense	Mistral 7b	66.1	55.8
StrategyQA	Commonsense	Llama 3.1 8b	73.7	68.6
StrategyQA	Commonsense	Llama 3.1 70b	85.3	83.8
StrategyQA	Commonsense	Gemma 2 9b	73.7	66.4
StrategyQA	Commonsense	Phi-3 Small 8k	72.3	66.0
StrategyQA	Commonsense	Qwen 2 7b	63.2	54.8
StrategyQA	Commonsense	Qwen 2 72b	81.7	76.9